

C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

API Documentation

Full REST API reference, 77 endpoints across 8 modules.

Program	Diploma in Advanced AI & Software Engineering
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Base URL: `http://127.0.0.1:8000/api` . All endpoints below are real, live routes taken directly from the running backend's OpenAPI schema — 77 in total. Authentication is a JWT bearer token unless marked Public.

1. Authentication & Users

Auth

ENDPOINT	AUTH	DESCRIPTION
POST /auth/register	Public	Register as customer or provider
POST /auth/login	Public	Returns a JWT bearer token
GET /auth/me	Token	Current user + provider profile if applicable

Users

ENDPOINT	AUTH	DESCRIPTION
GET /users/me	Token	Current user
PUT /users/me	Token	Update full_name, phone, password
GET /users/provider-profile	Provider	Own provider profile
PUT /users/provider-profile	Provider	Update bio, years_experience
GET /users/providers/{{provider_id}}	Public	Public provider profile

2. Services & Categories

Services

ENDPOINT	AUTH	DESCRIPTION
GET /services	Public	List active services
POST /services	Provider	Create a service
PUT /services/{{service_id}}	Provider	Update own service
DELETE /services/{{service_id}}	Provider	Remove own service
GET /services/mine	Provider	Own services
GET /services/categories	Public	List categories
POST /services/{{service_id}}/favourite	Customer	Save a favourite
DELETE /services/{{service_id}}/favourite	Customer	Remove a favourite
GET /services/favourites/me	Customer	Own favourites
POST /services/areas	Provider	Add a coverage area
GET /services/areas/mine	Provider	Own coverage areas
DELETE /services/areas/{{area_id}}	Provider	Remove a coverage area

3. Bookings & Availability

Bookings

ENDPOINT	AUTH	DESCRIPTION
POST /bookings	Customer	Request a booking
GET /bookings/customer	Customer	Own bookings
GET /bookings/provider	Provider	Own booking requests
GET /bookings/{{booking_id}}	Token	A single booking (participant or admin)
PUT /bookings/{{booking_id}}/accept	Provider	Accept a pending request
PUT /bookings/{{booking_id}}/reject	Provider	Reject a pending request
PUT /bookings/{{booking_id}}/cancel	Customer	Cancel a pending/accepted booking
PUT /bookings/{{booking_id}}/complete	Provider	Mark an accepted booking complete

Availability

ENDPOINT	AUTH	DESCRIPTION
POST /providers/availability	Provider	Add a weekly slot
GET /providers/availability/me	Provider	Own weekly slots
GET /providers/availability/provider/{{provider_id}}	Public	A provider's weekly slots
DELETE /providers/availability/{{slot_id}}	Provider	Remove a slot

4. Reviews & Payments

Reviews

ENDPOINT	AUTH	DESCRIPTION
POST /reviews/	Customer	Review a completed booking (once)
GET /reviews/	Public	List all reviews
GET /reviews/provider/{{provider_id}}	Public	Reviews for a provider
GET /reviews/service/{{service_id}}	Public	Reviews for a service
GET /reviews/customer/me	Customer	Own written reviews
GET /reviews/{{review_id}}	Public	A single review

Payments

ENDPOINT	AUTH	DESCRIPTION
POST /payments/	Customer	Pay for a completed booking
GET /payments/	Admin	All payments
GET /payments/booking/{{booking_id}}	Token	Payment for a booking
GET /payments/customer/me	Customer	Own payments
GET /payments/provider/me	Provider	Payments received
GET /payments/{{payment_id}}	Token	A single payment
PUT /payments/{{payment_id}}	Admin	Update status / transaction id

5. Notifications & Messages

Notifications

ENDPOINT	AUTH	DESCRIPTION
GET /notifications	Token	Own notifications
PUT /notifications/{{notification_id}}/read	Token	Mark one read

Messages

ENDPOINT	AUTH	DESCRIPTION
GET /messages/threads	Token	One thread per own booking, with unread count
GET /messages/bookings/{{booking_id}}	Token	Full message history; marks incoming as read
POST /messages/bookings/{{booking_id}}	Token	Send a message; notifies the other party

6. AI Assistant

AI

ENDPOINT	AUTH	DESCRIPTION
POST /ai/chat	Token	Conversational assistant; can check availability, create/cancel a booking
GET /ai/conversations	Token	Own conversation list
GET /ai/conversations/analysis	Token	Own chat usage stats
GET /ai/conversations/{{conversation_id}}	Token	Full message history for one conversation
POST /ai/recommend	Token	Category + real matching services from free text
POST /ai/provider-match	Token	Best-fit provider from a supplied shortlist
POST /ai/reviews/summarize	Provider	AI summary of a provider's own service reviews

The assistant's tools (`check_provider_availability` , `create_booking` , `cancel_booking`) call the exact same booking/availability logic as the REST endpoints above — there is one implementation of the business rules, not two. The same tools are also exposed to external MCP clients via a standalone MCP server (`backend/mcp_server/server.py`).

7. Admin

Admin

ENDPOINT	AUTH	DESCRIPTION
GET /admin/dashboard-stats	Admin	Platform totals: users, bookings, services, revenue, avg rating
GET /admin/revenue-timeseries	Admin	Real monthly revenue/booking chart data
GET /admin/users	Admin	Paginated user list
PUT /admin/users/{{user_id}}/status	Admin	Change account status
PUT /admin/users/{{user_id}}/email-verification	Admin	Toggle email verified
PUT /admin/users/{{user_id}}/provider-verification	Admin	Change provider verification status
GET /admin/categories	Admin	Categories with real service counts
POST /admin/categories	Admin	Create a category
DELETE /admin/categories/{{category_id}}	Admin	Remove an unused category
GET /admin/services	Admin	Every service, any status
PUT /admin/services/{{service_id}}/status	Admin	active / paused / draft
GET /admin/bookings	Admin	All bookings
GET /admin/reviews	Admin	Paginated review list
DELETE /admin/reviews/{{review_id}}	Admin	Remove a review
GET /admin/payments	Admin	All payments, with customer email
PUT /admin/payments/{{payment_id}}/status	Admin	Change payment status
POST /admin/seed-test-data	Admin	Idempotent demo data seeding (presentation aid only)

Health

ENDPOINT	AUTH	DESCRIPTION
GET /health	Public	Liveness check
GET /health/database	Public	Database connectivity check

8. Status Codes

CODE	MEANING
200 / 201	Success / Created
400	Business-rule violation (e.g. booking not completed, already reviewed)
401	Missing, invalid, or expired token
403	Wrong role, inactive account, or not a party to the resource
404	Resource not found
409	Duplicate email on registration
422	Validation error